



**Network Modeling and Control of
Physical Systems**

Mechanism Topology

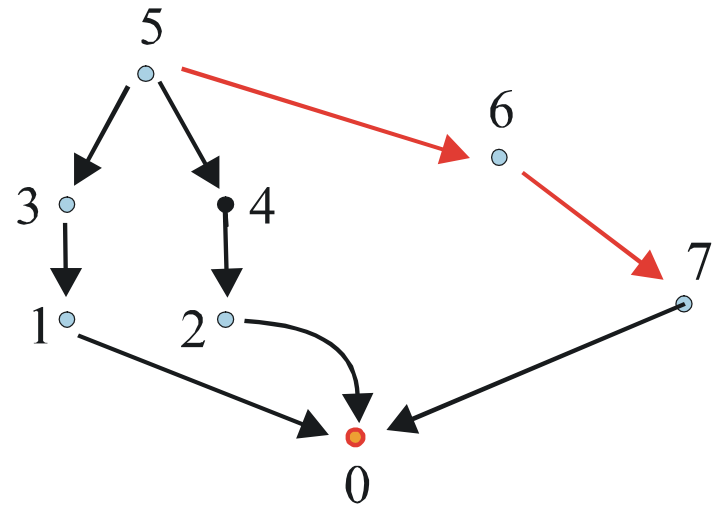
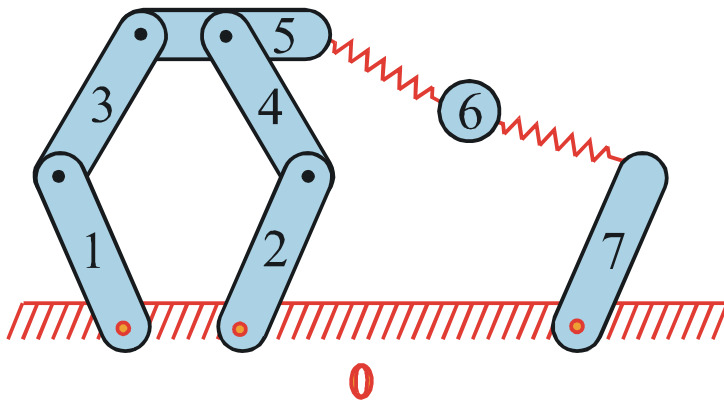
Network Topology

- Interconnection of q rigid bodies by n nodic elements (kinematic pairs, flexures, springs or dampers).
- We can define the **Primary Graph** describing the mechanism and than:

Port connection graph =

Lagrangian tree + Primary Graph

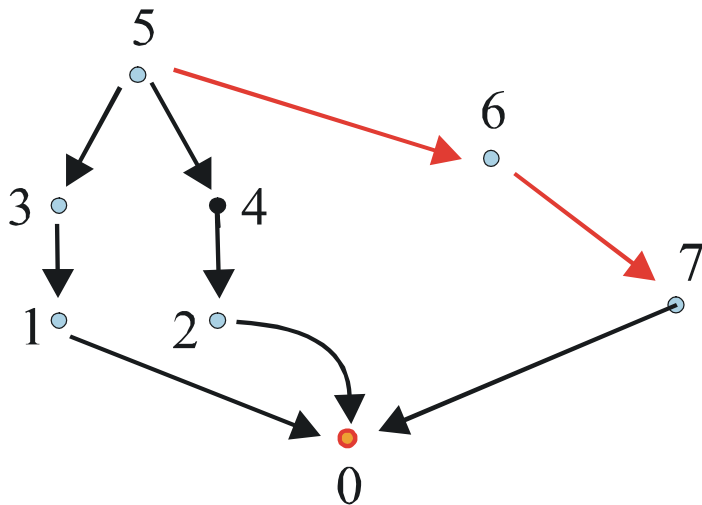
Primary Graph



Primary Graph

- The Primary graph is characterised by the **Incidence Matrix**

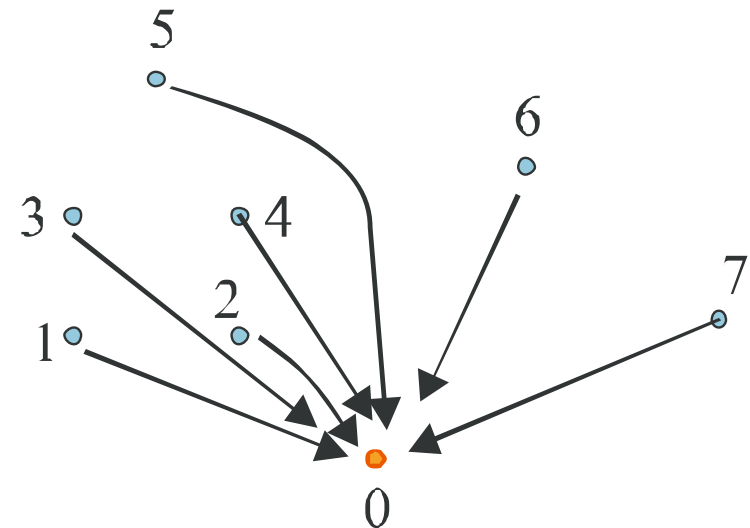
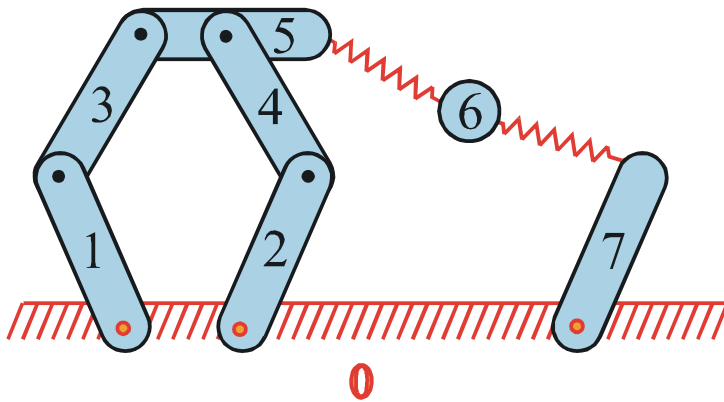
$$B \in \{-1, 0, 1\}^{q \times n}$$



$$n = 9$$

$$B = \begin{pmatrix} 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & -1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & -1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & -1 \\ 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix} q = 8$$

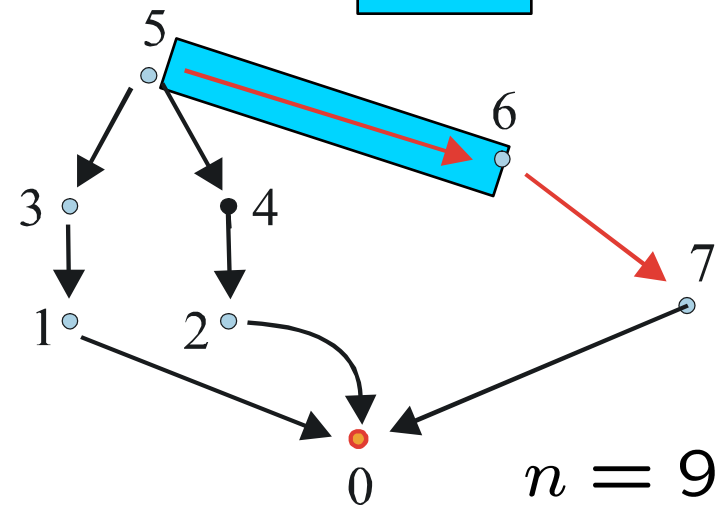
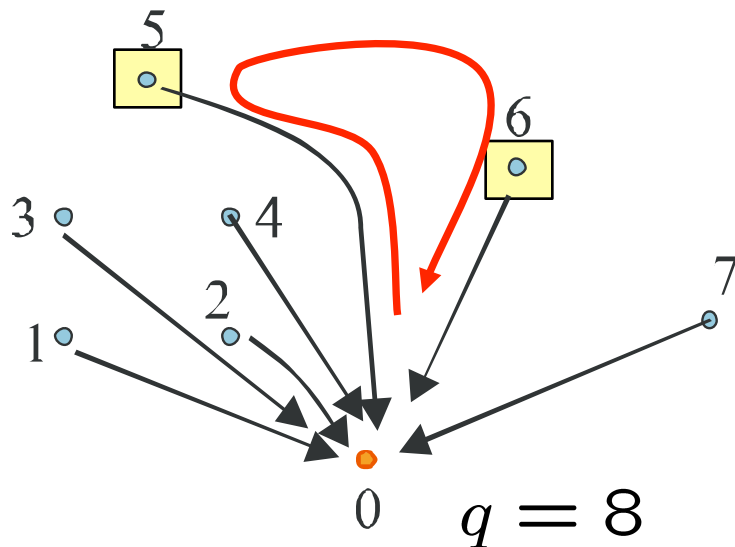
Lagrangian Tree



Fundamental Loop Matrix

$$C = \begin{pmatrix} B^T & I_{n \times n} \end{pmatrix} \in \mathbb{R}^{n \times (q+n)}$$

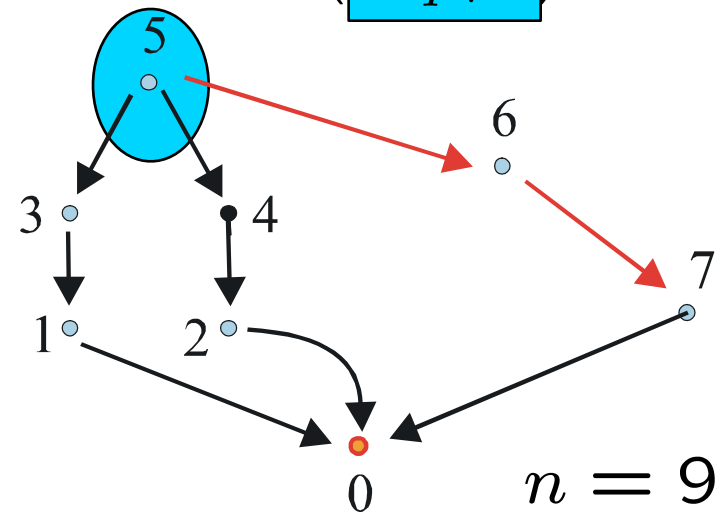
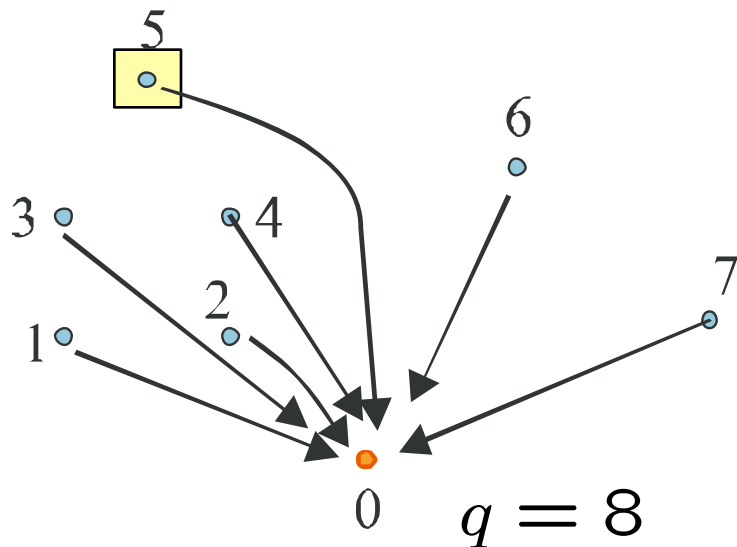
$$C \begin{pmatrix} T_1 \\ \vdots \\ T_q \\ T_{q+1} \\ \vdots \\ T_{q+n} \end{pmatrix} = 0$$



Fundamental Cut-set Matrix

$$Q = \begin{pmatrix} I_{q \times q} & -B \end{pmatrix} \in \mathbb{R}^{q \times (q+n)}$$

$$Q \begin{pmatrix} W_1 \\ \vdots \\ W_q \\ W_{q+1} \\ \vdots \\ W_{q+n} \end{pmatrix} = 0$$



Power Continuity

$$C = \begin{pmatrix} B^T & I_{n \times n} \end{pmatrix}$$

$$Q = \begin{pmatrix} I_{q \times q} & -B \end{pmatrix}$$



$$C Q^T = 0$$

**Power
continuity !**

$$\begin{pmatrix} C \\ 0_{q \times (q+n)} \end{pmatrix} \begin{pmatrix} T_1 \\ \vdots \\ T_q \\ T_{q+1} \\ \vdots \\ T_{q+n} \end{pmatrix} + \begin{pmatrix} 0_{n \times (q+n)} \\ Q \end{pmatrix} \begin{pmatrix} W_1 \\ \vdots \\ W_q \\ W_{q+1} \\ \vdots \\ W_{q+n} \end{pmatrix} = 0$$

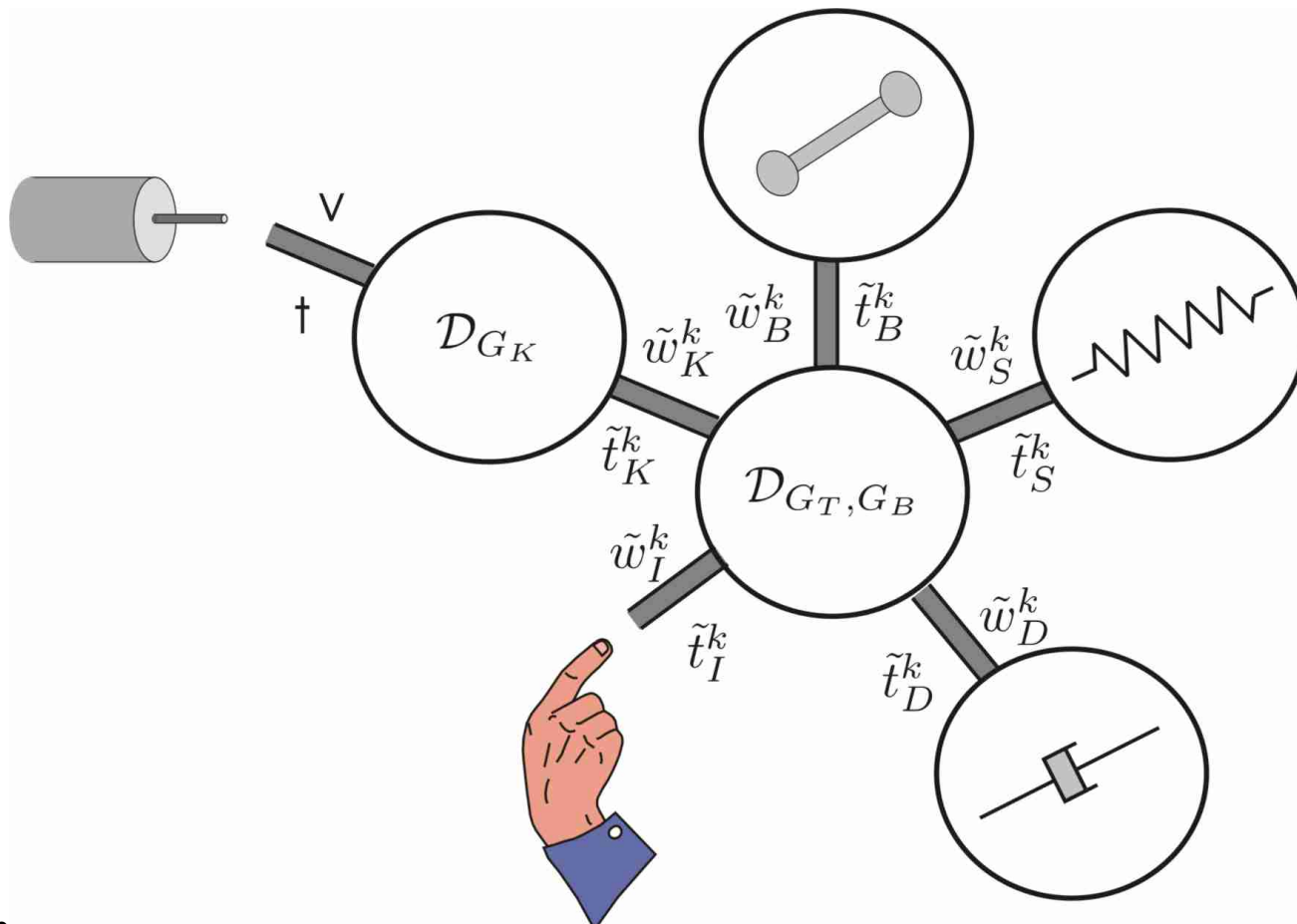
Mechanism Dirac Structure

Power Ports Rigid Bodies

$$\begin{pmatrix} C \\ 0_{q \times (q+n)} \end{pmatrix} \begin{pmatrix} T_1 \\ \vdots \\ T_q \\ T_{q+1} \\ \vdots \\ T_{q+n} \end{pmatrix} + \begin{pmatrix} 0_{n \times (q+n)} \\ Q \end{pmatrix} \begin{pmatrix} W_1 \\ \vdots \\ W_q \\ W_{q+1} \\ \vdots \\ W_{q+n} \end{pmatrix} = 0$$

Power Ports Nodic Elements

Further Steps...



Conclusions

- Any 3D part can be modeled in the Dirac framework
- Any interconnection also !
- In this case the ports have a geometrical structures: no scalars !
- Some steps still to go to bring the system in explicit form
- A lot of extensions are possible
- Not trivial to bring everything in simplified explicit form

next